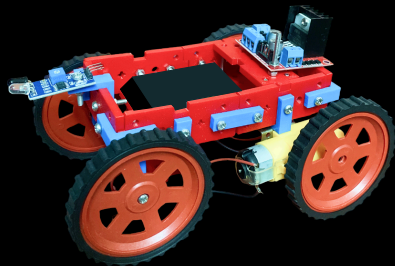


level 2

no. of sessions: 5

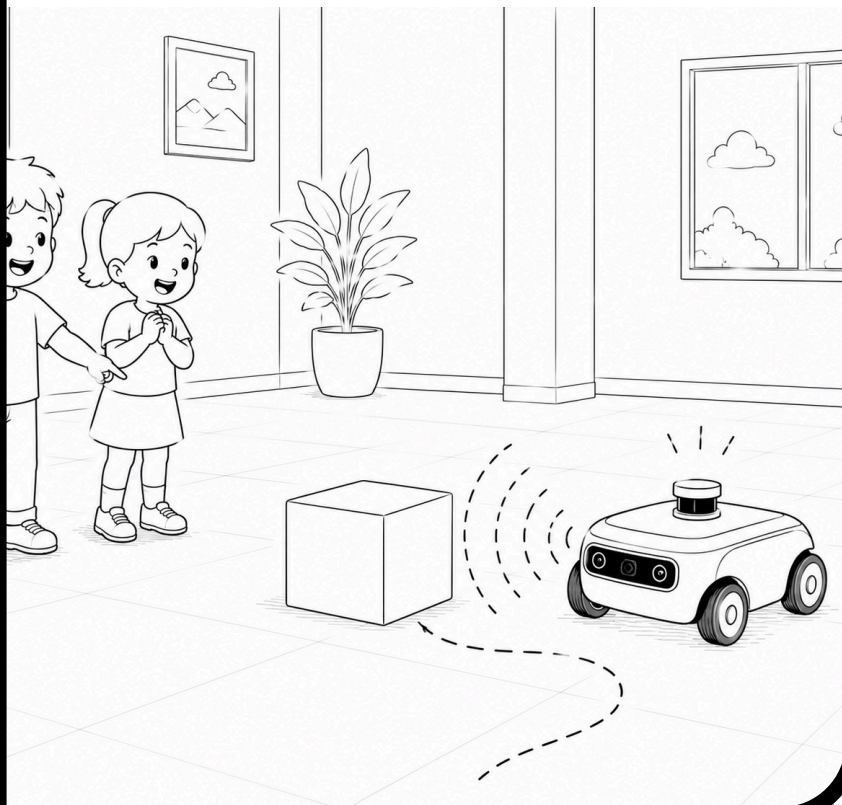
obstacle avoider bot

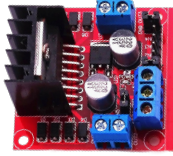
The obstacle avoider uses sensors to detect objects and stops to avoid crashing, just like a robot vacuum avoiding furniture.



simple machine: wheels
electronic parts: input vs output,
current & voltage

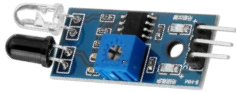
oh.





Motor Driver

The **brain of a robot**. I listen to input devices like sensors and tell output devices like motors/LEDs when to turn ON/OFF.



IR Sensor

I am like a **robot's eyes**. I help robots see objects or follow a path without touching anything.



LDR Sensor

I am like a **light detector** who tells a robot if it is bright or dark and i am used in automatic lights.



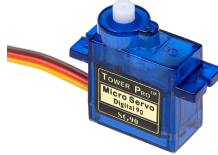
Motor

I **change electricity into movement** and help wheels, fans, and robots spin and move.



DPDT Switch Controller

Like a **direction button** - i make a motor spin forward or backward without using coding.



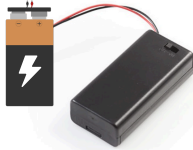
Servo Motor

I am a **smart motor** that can stop at the exact position you want. I am used in robot arms, gates, and moving parts.



Servo Controller

I am like a **control room** for the servo motor and lets you move the servo easily to test how it works.



Battery & Battery Holder

I **give power** to the project, and my battery holder keeps the battery safe and connected. Together, we help the model work.



Jumper Wires

We are the **roads for electricity** who connect the different parts of the circuit, so power can travel.



Motor Clamp

I am like a **strong hand** that holds the motor tightly in place while it spins.

materials needed



M-F beam - x6



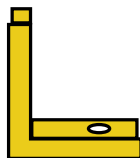
small screw - x12



nut - x12

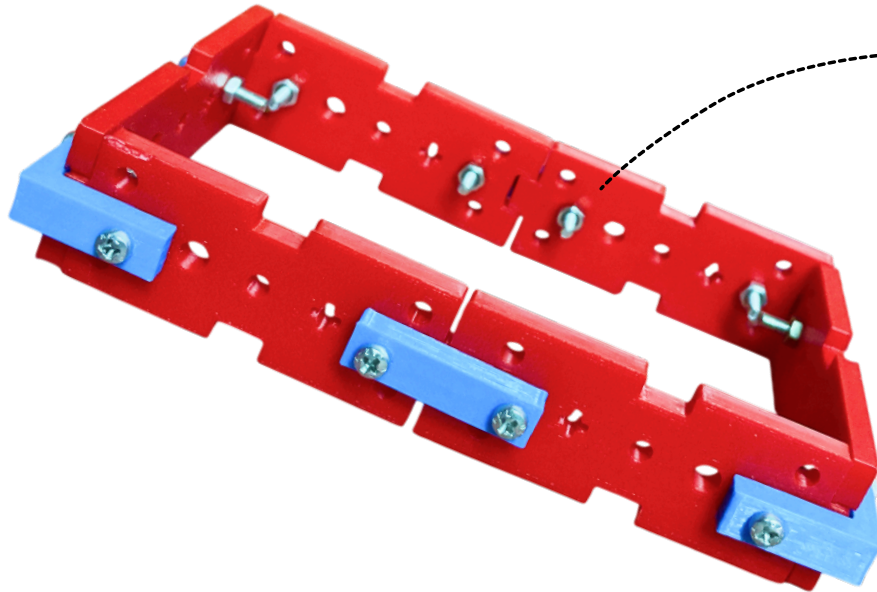


I clamp - x2



L clamp - x4

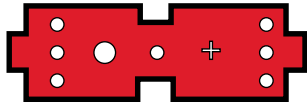
step 1: assemble the body



Join **6 M-F beams** together to form a rectangle shape.

Use **L and I clamps** to hold them together.

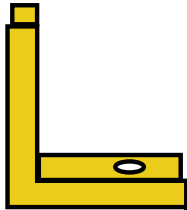
materials needed



M-M beam - x2



motor clamp - x2



L clamp - x4



long screw - x4



small screw - x4



nut - x10

step 2: assemble the motor supports

1.

Attach **2 M-M beams** to the rectangle shape

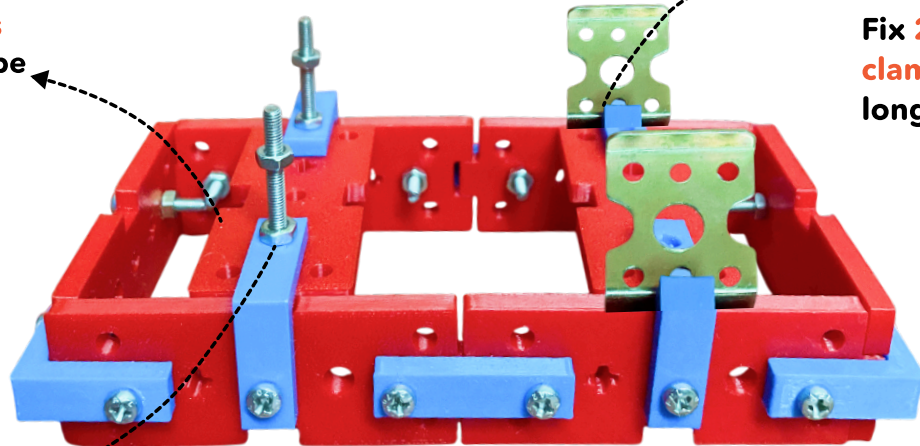
Use **L clamps** to hold them together

3.

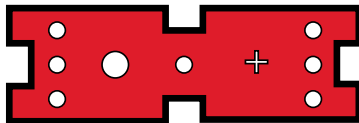
Fix **2 motor clamps** using long screws

2.

Fix **2 long screws** only to this M-M beam



materials needed

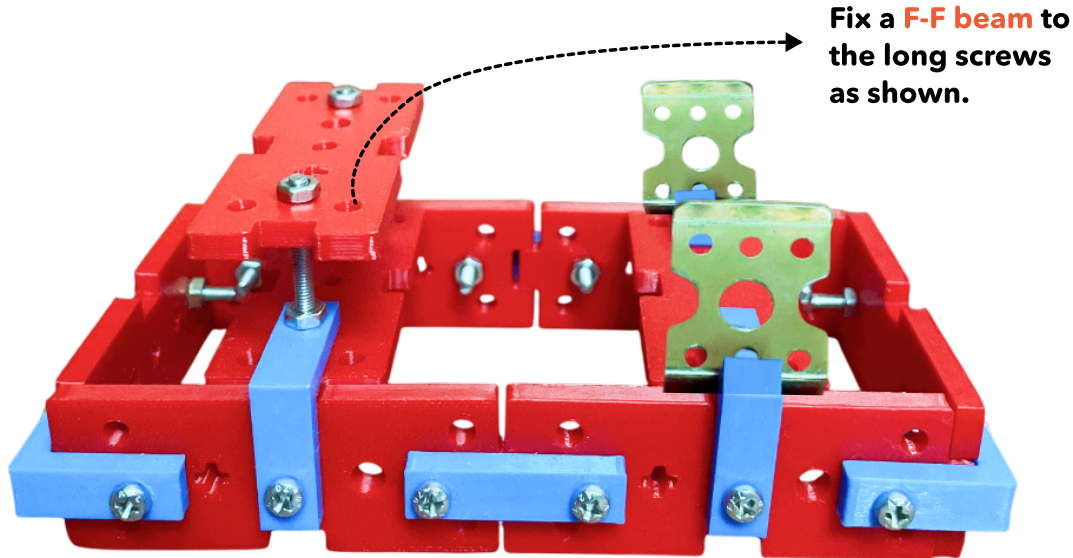


F-F beam - x1



nut - x2

step 3: assemble the battery support



materials needed



I clamp - x2

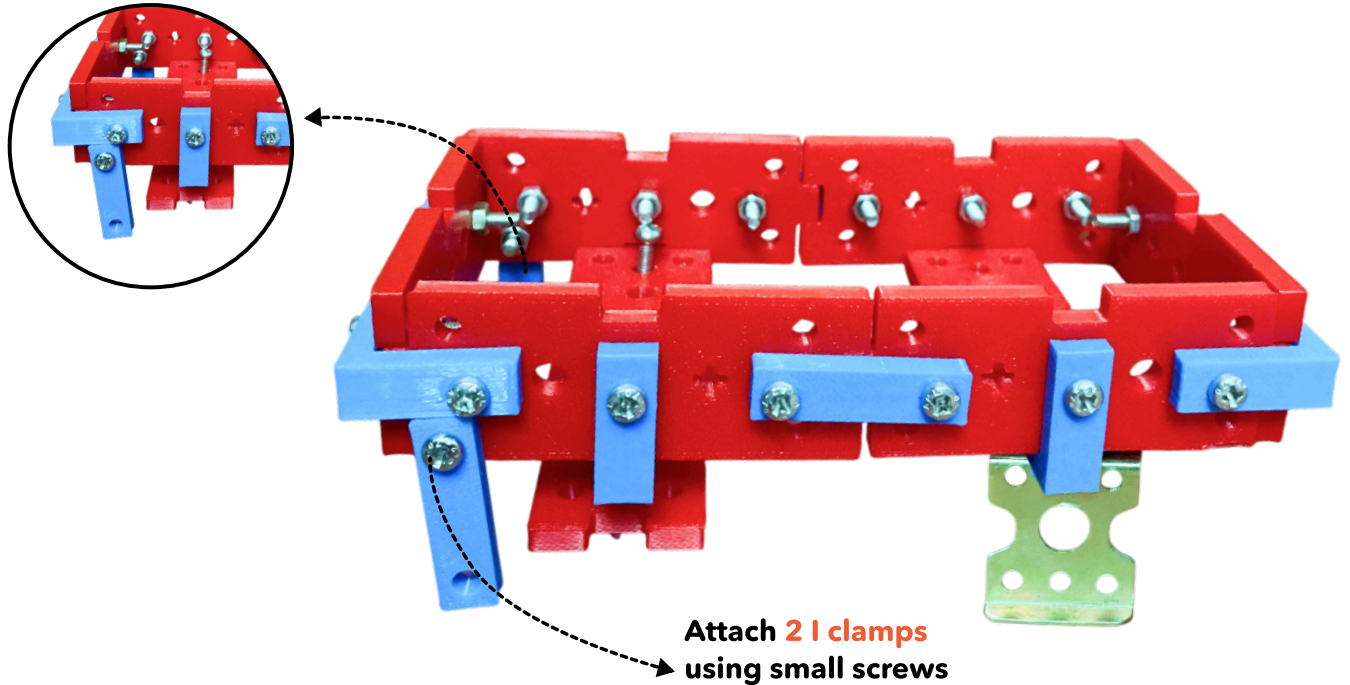


small screw - x2



nut - x2

step 4: assemble the wheel support



materials needed



motor - x2

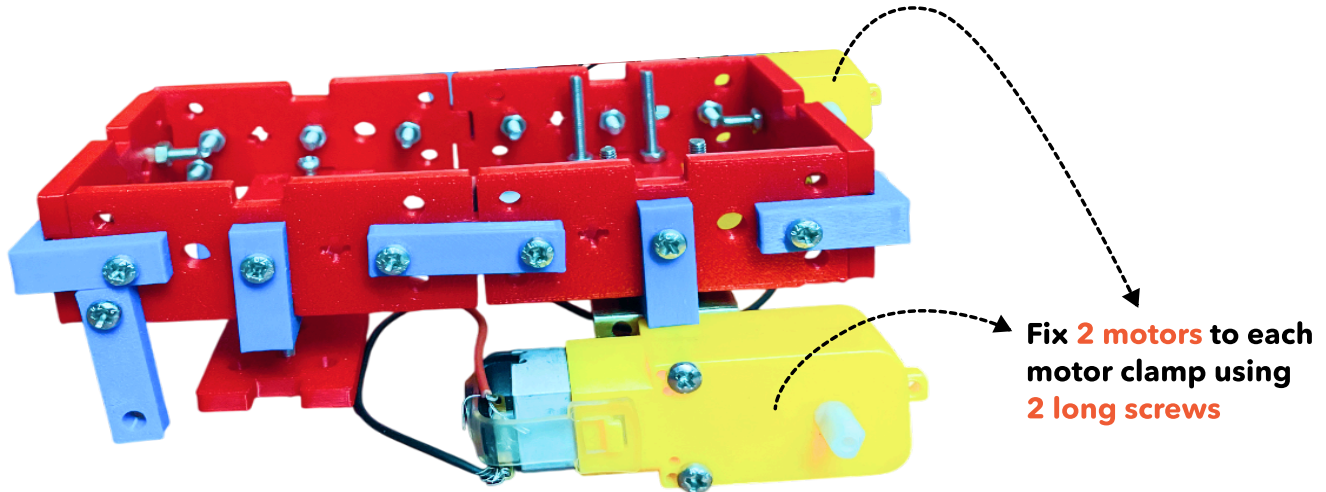


long screw - x4



nut - x4

step 6: fix the motors



materials needed



wheel - x4



long screw - x2

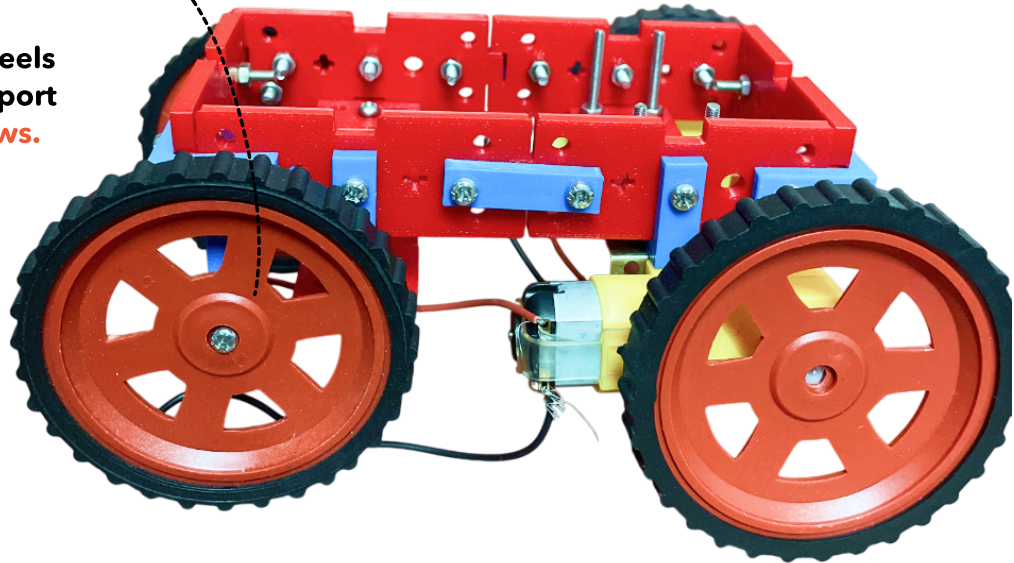


nut - x2

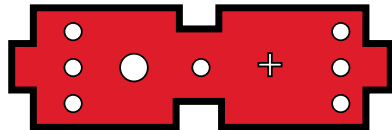
step 7: fix the wheels

Attach the **wheels** to the motors axle as shown.

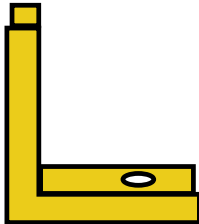
Then fix the wheels to the front support using **long screws**.



materials needed



m-m beam - x1



L clamp - x1



long screw - x1



small screw - x1



nut - x2

step 8: add the motor driver support

1.

Attach a **long screw** to the a **MM beam**



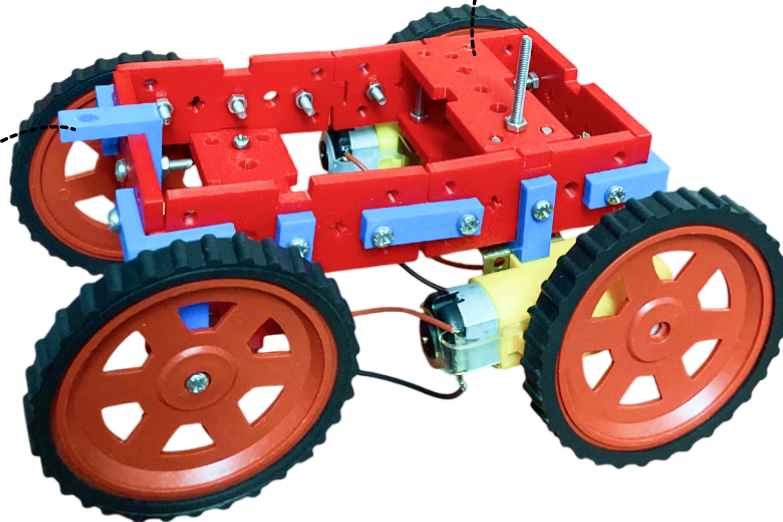
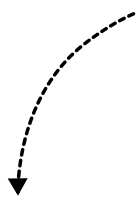
2.

Fix the **M-M beam** without any screws

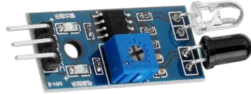


3.

Fix the **L clamp** to the front



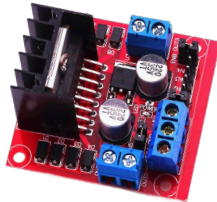
materials needed



IR sensor x1



battery holder x1



motor driver x1

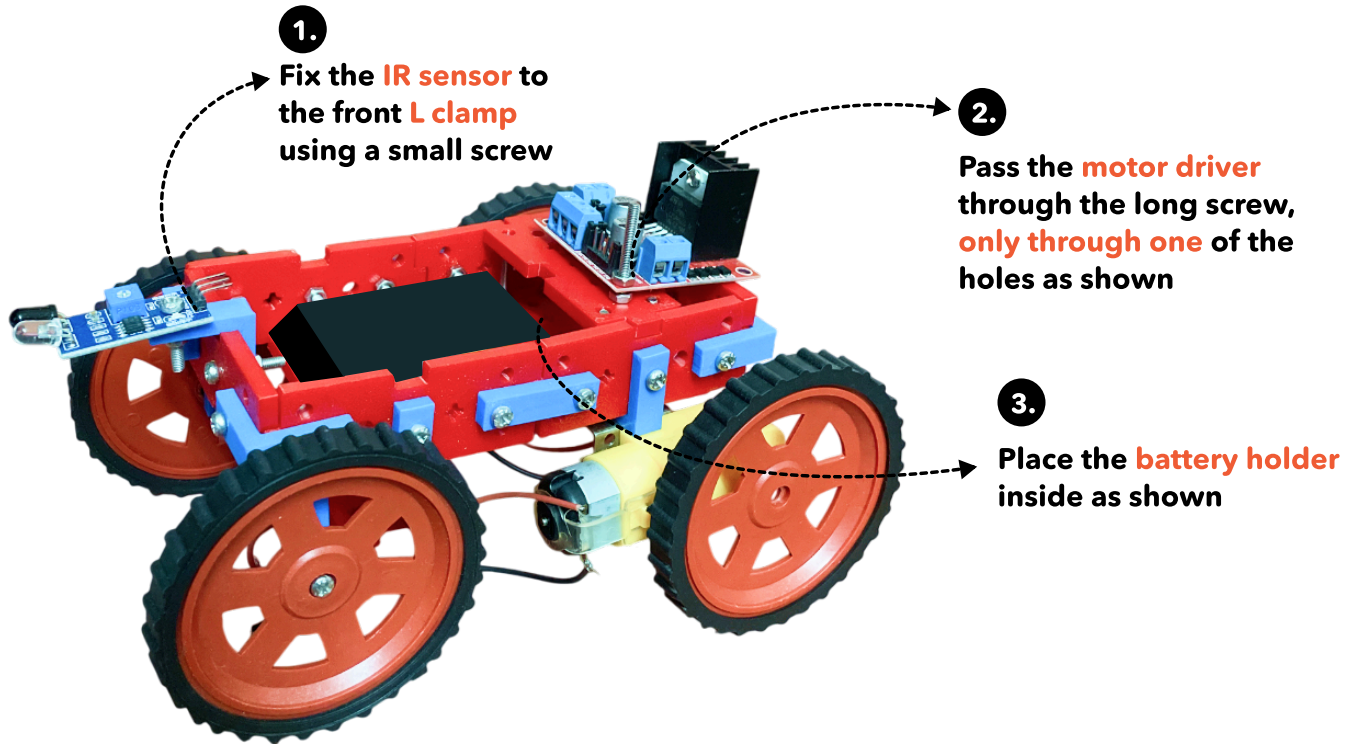


small screw - x1



nut - x2

step 9: add the motor driver and sensor



materials needed



battery holder x1



battery x1



jumper cables - X1

step 10: circuit and testing

